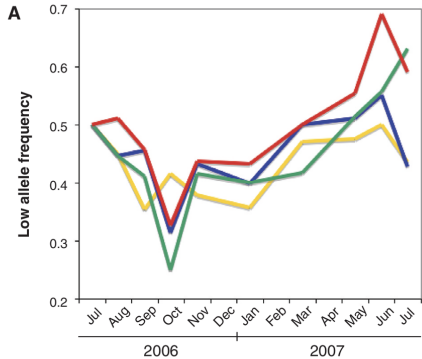


Natural selection in Threespine sticklebacks



Selection at a biallelic locus

link: <https://phytools.shinyapps.io/selection/>

Selection at a biallelic locus

p0 (initial value of p):

0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1

number of generations:

0 203 1,000

0 100 200 300 400 500 600 700 800 900 1,000

Show plot of:

p

Fitnesses:

W(AA)	W(Aa)	W(aa)
1.00	1.00	0.80

☐ show equilibrium (if applicable)

☒ reactive (plot changes instantly with new parameter values)

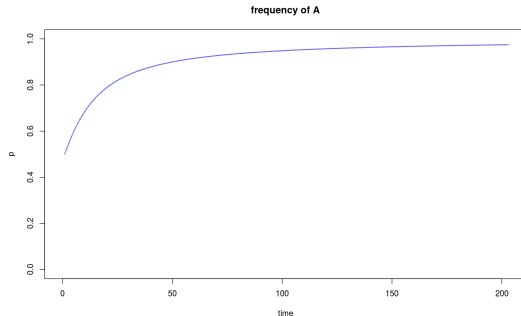
new plot

Instructions:

This application numerically analyzes a simple natural selection model for a biallelic locus.

Details:

Web interface for selection of the learnPopGen R package (Revell, 2019).



Selection Activity

- 1) Set fitnesses to $W(AA)=1.0$, $W(Aa)=1.0$, and $W(aa)=0.8$. Set p_0 to 0.01, and make sure you are showing a plot of p . What mode of inheritance is occurring at this locus? What is the value of the selection coefficient? How many generations does it take to reach $p=0.6$?
- 2) Change $W(Aa)$ to 0.9. What mode of inheritance are you modeling now? Does the time it take to reach $p=0.6$ change?
- 3) Switch to showing a plot of q without changing any other settings. What can you conclude about the relationship between p and q through time?
- 4) Develop a question and address it using this app. Be prepared to report back to the class.

Nachman et al. Group Discussion

- 1) What is the mechanism of natural selection in this study?
- 2) What does it mean that MC1R is “perfectly associated” with coat color?
- 3) What does Table 2 tell us?
- 4) What does Figure 2 tell us?